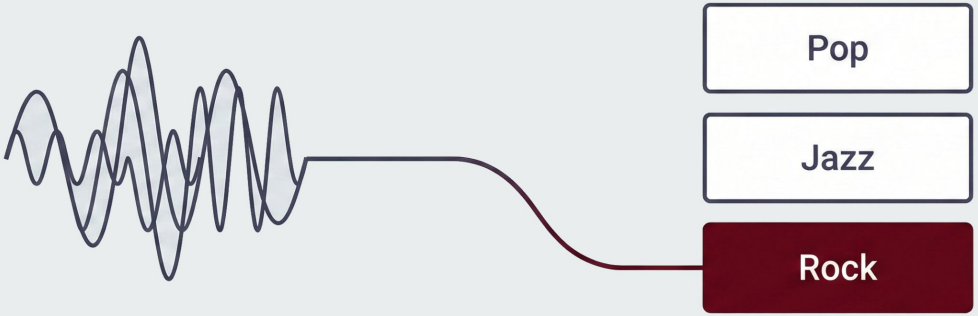
Music genre classification



Foundation of Data Science
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Sapienza University of Rome

Mariotti - 1936936
Pizzari - 1936451
Romito - 1932500

Task and motivation

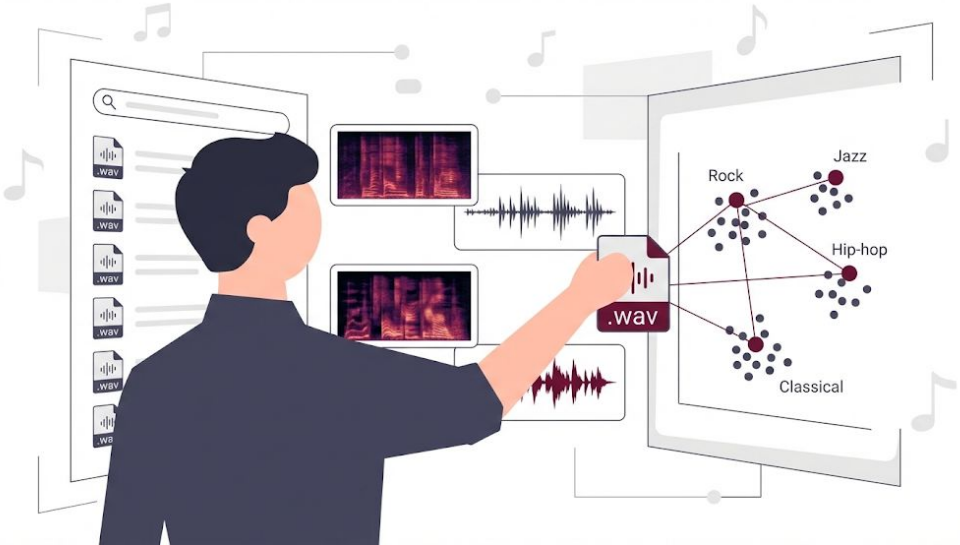
Genre classification is a crucial task in a world where the prevalent way to consume music is in digital form.

Automatic genre classification allows to:

- Organize songs efficiently
- Provide precise recommendation systems
- Support musicological research
- Enable semantic search across large audio databases
- And more...

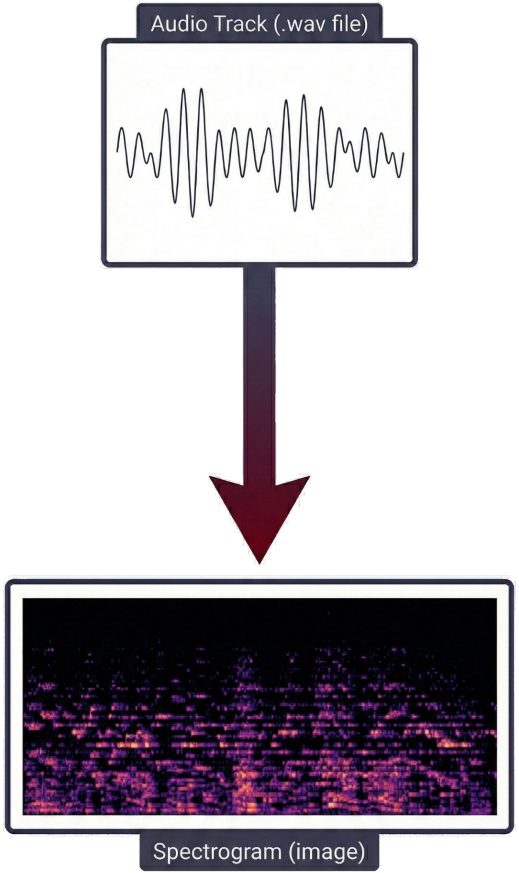
GTZAN Dataset - Music Genre Classification

- **Pioneering Dataset**: The GTZAN dataset is a foundational benchmark in Music Genre Recognition (MGR) research. It is often referred to as the "MNIST of sounds."
- **Content:** It contains 1,000 audio tracks (30 seconds each) categorized into 10 distinct music genres (100 tracks per genre).
- **Genres**: Blues, Classical, Country, Disco, Hip-Hop, Jazz, Metal, Pop, Reggae and Rock.



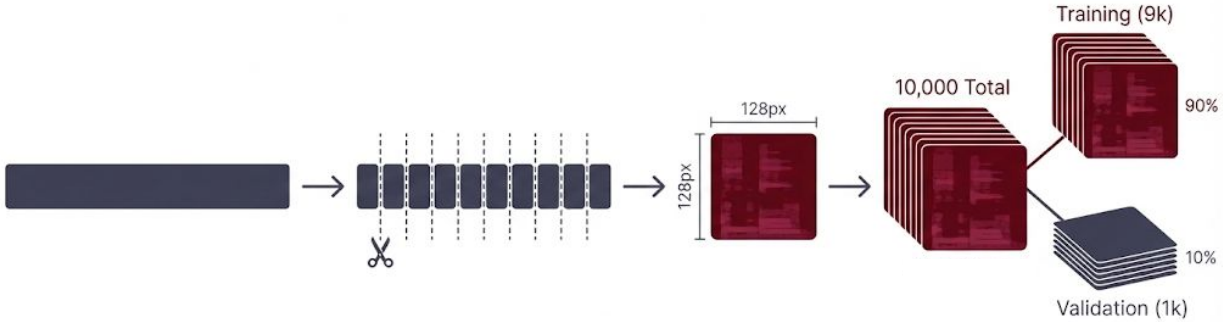
Audio track to spectrogram conversion

- **A "heatmap" of sound:** visual representation of the spectrum of frequencies of a signal as it varies with time.
- **X-Axis**: Time (how long the sound lasts).
- **Y-Axis**: Frequency (low bass at the bottom, high treble at the top).
- **Color/Brightness**: Sound intensity (brighter colors = louder sounds).
- Allows to take a **computer vision approach** to solve our task



Data preparation

- **Segmentation:** Tracks were sliced into 3-second clips (10 segments per track).
- **Input Dimensions:** Converted to 128x128 pixel spectrogram images.
- **Dataset Size:** 10,000 total segments (balanced at 1,000 per genre).
- **Train/Val Split:** 90% Training (9,000) vs. 10% Validation (1,000).



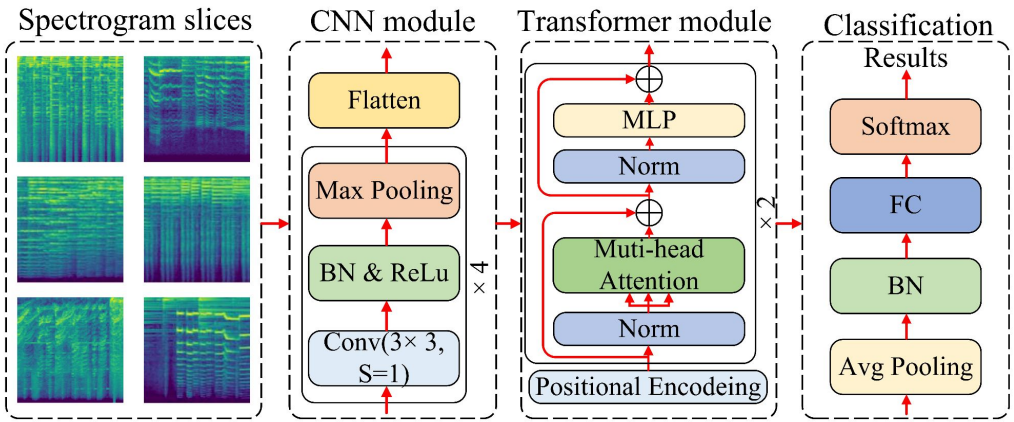
Full Audio Track

Segmentation:
3s clips (10 per track)

Input: 128x128
Spectrograms

Split:
90% Train / 10% Val

The model - First implementation

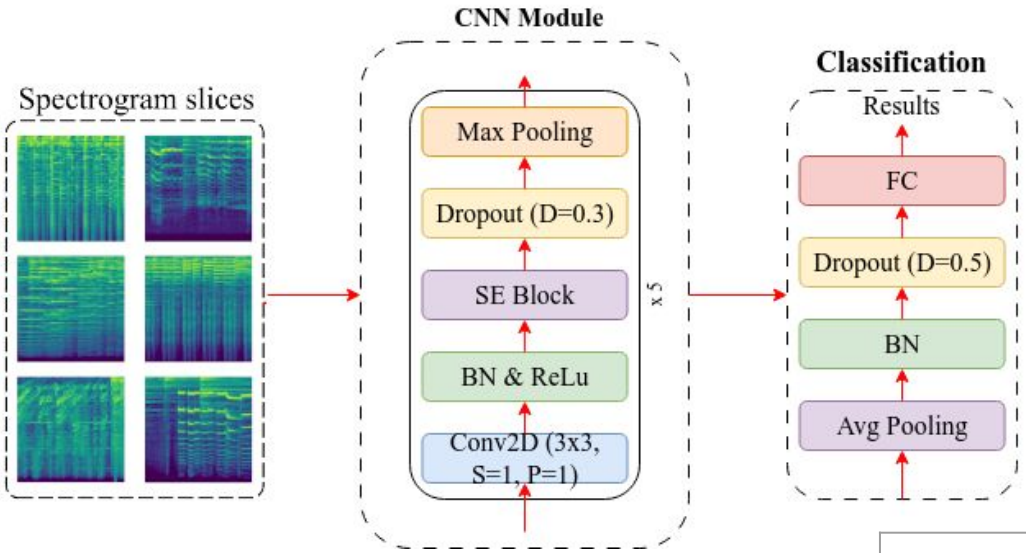


After many attempts, we were not able to achieve results as high as those described in the paper.

The best run achieved:

Accuracy	F1	Recall	Precision	Epochs (per fold)
79,62%	78,38%	79,66%	78,63%	100

The model - Our implementation



Let's simplify and try to improve attention directly on the CNN module

The best run achieved:

Accuracy	F1	Recall	Precision	Epochs (per fold)
85,14%	83,86%	85,12%	85,45%	100

Data augmentation - Audiomentations package

Techniques Used

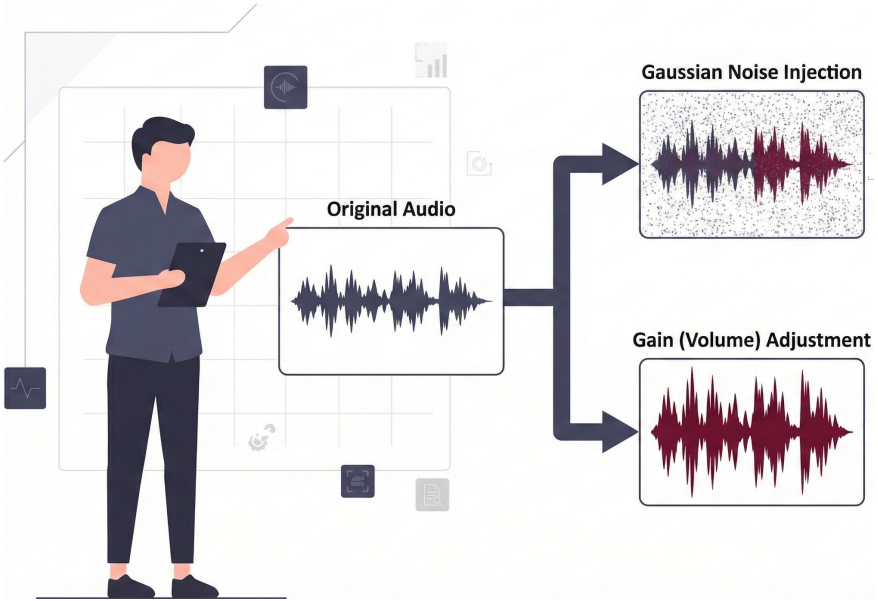
- **Gaussian Noise:** Adds noise to mimic recording imperfections.
- **Gain:** Randomizes volume (dB) to vary loudness.

Execution Logic

- **1 Input → 1 Output** per call.
- Run multiple times to generate distinct variations.

Results

- **Accuracy:** Remained close to baseline runs (without augmentation).
- **Impact:** No significant improvement observed, so this techniques were discarded



...and now a live demo!

Conclusions

Model Evolution & Results

- **Initial Approach:** Hybrid CNN-Transformer model fell short of expectations (~79.6% accuracy).
- **Winning Architecture:** The optimized CNN with SE Blocks and dropout achieved the best performance, outperforming the starting hybrid structures.
- **Final Accuracy:** Reached a peak of **85.14%**.

Future Improvements

- Calculate the BPM and the tonality of a track in addition to the genre.
- Identify the instrument used in the track to try to guess the artist.

Thank you for the attention!

- **Dataset:**
  - https://www.kaggle.com/datasets/andradaolteanu/gtzan-dataset-music-genre-classification
    G. Tzanetakis and P. Cook, "Musical genre classification of audio signals," in IEEE Transactions on Speech and Audio Processing, vol. 10, no. 5, pp. 293-302, July 2002, doi: 10.1109/TSA.2002.800560.}
- **Papers:**
  - https://www.mdpi.com/2079-9292/13/16/3313
    A hybrid parallel computing architecture based on CNN and Transformer for music genre classification (Chen, Jiyang and Ma, Xiaohong and Li, Shikuan and Ma, Sile and Zhang, Zhizheng and Ma, Xiaojing)
- **Libraries:**
  - https://github.com/iver56/audiomentations
    Audiomentations: *a Python library for audio data augmentation. Useful for making audio ML models work well in the real world, not just in the lab.*
- **Slides:**
  - https://github.com/pietro-nardelli/sapienza-ppt-template
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